

Feedback — Lecture 5 Quiz

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You submitted this quiz on **Thu 23 Jul 2015 8:04 AM PDT**. You got a score of **10.00** out of **10.00**.

Question 1

A student writes several functions to swap the values of the two variables x and y, i.e. if x=1 and y=2, after calling the swap function x=2 and y=1. The different functions that the student writes are given below: def swap1(x,y) :

```
x=y
y=x
return(x,y)
```

```
def swap2(x,y) :
    return(y,x)
```

```
def swap3(x,y) :
    z=x
    x=y
    y=z
    return(x,y)
```

```
def swap4(x,y) :
    x,y=y,x
    return(x,y)
```

Which of the functions swap1, swap2, swap3, and swap4 is correct?

Your Answer	Score	Explanation
☐ None		
☒ Functions swap2, swap3, and swap4 only	✓ 1.00	
☐ Functions swap1, and swap3 only		
☐ Functions swap1, and swap2 only		

Total

1.00 / 1.00

Question 2

Consider the following two functions:

```
def f1(x):
```

```
    if (x > 0):
```

```
        x = 3*x
```

```
        x = x / 2
```

```
    return x
```

```
def f2(x):
```

```
    if (x > 0):
```

```
        x = 3*x
```

```
    x = x / 2
```

```
    return x
```

For what values of x will f1 and f2 return the same value?

Your Answer	Score	Explanation
☒ When x is zero or positive	✓ 1.00	
☐ For positive values of x only		
☐ For negative values of x only		
☐ For x		
Total	1.00 / 1.00	

Question 3

A recursive function in programming is a function that calls itself during its execution. The following two functions are examples of recursive functions:

```
def function1(length):
```

```
    if length > 0:
```

```

    print(length)
    function1(length - 1)
def function2(length):
    while length > 0:
        print(length)
        function2(length - 1)

```

What can you say about the output of function1(3) and function2(3)?

Your Answer	Score	Explanation
☐ The two functions produce the same output 1 2 3.		
☐ The two functions produce the same output: 3 2 1		
☐ function1 produces the output: 3 2 1 and function2 runs infinitely.		
☒ function1 produces the output: 3 2 1 and function2 runs infinitely.	✓ 1.00	
Total	1.00 / 1.00	

Question 4

The following recursive function takes three positive integer arguments:

```

def compute(n,x,y)
    if n==0 : return x
    return compute(n-1,x+y,y)

```

What is the value returned by the compute function?

Your Answer	Score	Explanation
☐ n*x+y		
☒ x+n*y	✓ 1.00	
☐ x+y		

☐ $n \cdot (x+y)$

Total

1.00 / 1.00

Question 5

What will the returned value be for the compute function defined in Exercise 3 if the argument n is negative?

Your Answer

Score

Explanation

☐ $x+n \cdot y$

☒ The function will never return a value.



1.00

☐ $n \cdot x + y$

☐ x

Total

1.00 / 1.00

Question 6

The following functions are all intended to check whether a string representing a dna sequence contains any characters that are not 'a','c','g','t', 'A', 'C', 'G', or 'T'. At least some of these functions are wrong. Which ones are correct?

```
def valid_dna1(dna):
```

```
    for c in dna:
```

```
        if c in 'acgtACGT':
```

```
            return True
```

```
        else:
```

```
            return False
```

```
def valid_dna2(dna):
```

```
    for c in dna:
```

```
        if 'c' in 'acgtACGT':
```

```
            return 'True'
```

```
        else:
```

```
return 'False'
```

```
def valid_dna3(dna):
    for c in dna:
        flag = c in 'acgtACGT'
    return flag
```

```
def valid_dna4(dna):
    for c in dna:
        if not c in 'acgtACGT':
            return False
    return True
```

Your Answer	Score	Explanation
☒ valid_dna4 only	✓ 1.00	
☐ None of them.		
☐ valid_dna2, and valid_dna4 only		
☐ valid_dna1, valid_dna2, and valid_dna3 only		
Total	1.00 / 1.00	

Question 7

What is the type of variable L3 and what is its value if L1 and L2 are lists?

L3 = [i for i in set(L1) if i in L2]

Your Answer	Score	Explanation
☐ L3 is a set with elements common between the lists L2 and L3.		
☐ L3 is a tuple with elements that are both in L1 and L2		
☐ L3 is a list with all the elements in L1 and L2		
☒ L3 is a list that contains only the elements that are common between the lists (without duplicates).	✓ 1.00	
Total	1.00 / 1.00	

Question 8

What will be printed after executing the following code?

```
>>>def f(mystring):  
    print(message)  
    print(mystring)  
    message="Inside function now!"  
    print(message)  
  
>>>message="Outside function!"  
  
>>>f("Test function:")
```

Your Answer	Score	Explanation
☒ An error message.	✓ 1.00	
☐ Inside function!		
☐ Outside function!		
☐ Test function: Inside function now!		
Total	1.00 / 1.00	

Question 9

Which statement below is true about a function:

Your Answer	Score	Explanation
☐ must have at least one parameter		
☐ its arguments always appear within brackets		
☐ must always have a return statement		
☒ may have no parameters	✓ 1.00	
Total	1.00 / 1.00	

Question 10

Which of the following function headers is correct?

- A. `def afunction(a1 = 1, a2):`
- B. `def afunction(a1 = 1, a2, a3 = 3):`
- C. `def afunction(a1 = 1, a2 = 2, a3 = 3):`

Your Answer	Score	Explanation
☒ C	✓ 1.00	
☐ A,C		
☐ B,C		
☐ A,B,C		
Total	1.00 / 1.00	